Workpackage X.X:

Calculation Engine associated with EFI+ indices: R application 'efitools-1.1' (beta version)

In construction!



Authors:

Pierre Bady* pierre.bady@cemagref.fr
Maxime Logez** maxime.logez@cemagref.fr
Didier Pont* didier.pont@cemagref.fr

* CEMAGREF (HBAN), Parc de Tourvoie, BP 44, 92163 Antony Cedex, FRANCE
** CEMAGREF (HYAX), 3275 Route de Cézanne, CS 40061, 13182 AIX EN PROVENCE Cedex 5, FRANCE

Created: 01/09/2008
Revision: 18/03/2009
Version: 1.0

Abstract:

This document contains the main information on the development of the computational engine associated with EFI+ indices.

Keywords: EFI+ project, calculation engine, R, efitools, fish index

Contents:

1- Motivations 2
2- Programming Tools 2
3- R application 2
3.1- Application Design 2
3.2- Installation 6
3.3- How to use R-script to compute EFI index 6
3.4- Description of the R application (file main.R or maindiad.R) 7
3.5- Output file 7
3.6- Additional Information and limitation 9
4- Reference 10

1- Motivations

This document contains the main elements on the input and output files associated with the computation of the EFI+ index. We quickly present a simplified design and organisation of the procedure in the ***Figure 1*** and we describe the most important points for each step of the procedure.

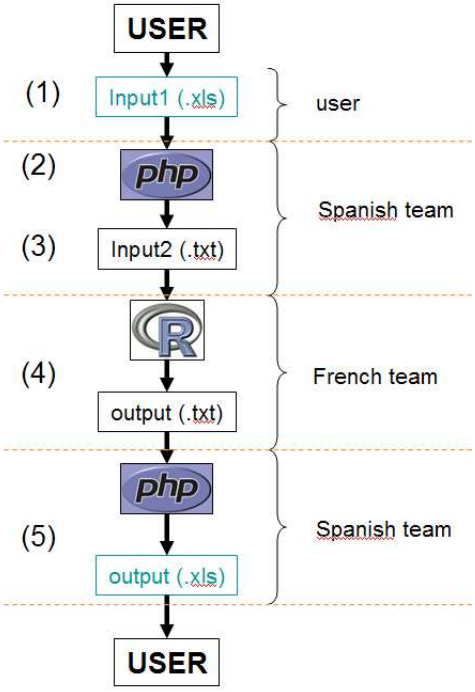


Figure 1. Simplified design and organisation.

2- Programming Tools

For more detail concerning the statistical methods, we encouraged people to consult the appropriate bibliography. All the routines necessary for computing models were implemented in the R[1]software version > **2.6.0** (R Development Core Team 2007). The new functions only depend to the base packages (e.g. base, stats, MASS).

3- R application

3.1- Application Design

The global organisation of the dataset (Input2) must be similar to the structure of the User input1 data file. In the Table 1, we indicate the minimal list of necessary variables for the computation of the

[1] R Development Core Team (2007). R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. ISBN 3-900051-07-0, URL http://www.R-project.org.

expected values and scores values. If there are additional variables, it's not a real problem, because we use particular filters to only select valid variables (based on the names of the variables).

Table 1. Description and structure of the exchange file between PHP application and R. The change with the previous version is highlighted in Yellow.

variables	**type**	**Description**
Sample code	Character	Site name (unique)
Day	Integer	e.g.: 08
Month	Integer	e.g.: 10
Year	Integer	e.g.: 2008
Longitude	Float	Longitude (in decimal degree)
Latitude	Float	Latitude (in decimal degree)
Actual.river.slope	Float	Slope of the river segment (in m/km)
Temp.jul	Float	July temperature (in °C)
Temp.jan	Float	January temperature (in °C)
Floodplain.site	Character	Presence or absence of floodpain (2 modalities, Yes or No)
Water.source.type	Character	Water source type (4 modalities, more description in database manual)
Geomorph.river.type	Character	(5 modalities, more description in database manual)
Distance.from.source	Float	Distance from the source (in km)
Area.ctch	Float	Upstream drainage area (in km2)
Natural.sediment	character	Natural sediment (5 modalities)
Ecoreg	Character	Illies ecoregions
Fished area	Float	In meter squared
Method	Character	Wading or Boating or Mixed
Sampling.location	Character	'Backwaters', 'Mixed' or 'Main channel'
Eft.type	Character	Functional typology proposed on the FAME project (see FAME documentation and Andy for more details).
Medit	Integer	Variable which indicates the mediterranean status of the site.
Species	Character	Species names of the fish
Total.number.run1	Float	Fish number for a given species for the first Run (see database manual for more details).
Number.length.below.150	Float	Fish number **inferior or equal to 150mm** for a given species for the first Run (see database manual for more details).
Number.length.over.150	Float	Fish number **superior to 150mm** for a given species for the first Run (see database manual for more details).

In the file 'input2 A' (Table 1), there is one line **by** fishing occasion **by** species name. In addition, an important point is the coding of the missing values: for character type, we use the term 'NoData' and we use the term '-999' for numerical fields (integer and float type). For more details, you can consult the documentation related to the EFI+ database.

Table 2 is related to the diadromous species scores computation input (contained in the file called 'diadromous.txt' in the R application).

Table 2. Description and structure of the exchange file between PHP application and R (diadromous species)

variables	type	Description
Sample code	Character	unique
Day	Integer	e.g.: 08
Month	Integer	e.g.: 10
Year	Integer	e.g.: 2008
Longitude	Float	Longitude (in decimal degree)
Latitude	Float	Latitude (in decimal degree)
Acipenser naccarii historic	integer	"-999" or "0" or "1"
Acipenser naccarii present	integer	"-999" or "0" or "1"
Acipenser gueldenstaedtii historic	integer	"-999" or "0" or "1"
Acipenser gueldenstaedtii present	integer	"-999" or "0" or "1"
Acipenser nudiventris historic	integer	"-999" or "0" or "1"
Acipenser nudiventris present	integer	"-999" or "0" or "1"
Acipenser stellatus historic	integer	"-999" or "0" or "1"
Acipenser stellatus present	integer	"-999" or "0" or "1"
Acipenser sturio/A. oxyrinchus historic	integer	"-999" or "0" or "1"
Acipenser sturio/A. oxyrinchus present	integer	"-999" or "0" or "1"
Alosa alosa historic	integer	"-999" or "0" or "1"
Alosa alosa present	integer	"-999" or "0" or "1"
Alosa fallax historic	integer	"-999" or "0" or "1"
Alosa fallax present	integer	"-999" or "0" or "1"
Alosa immaculata historic	integer	"-999" or "0" or "1"
Alosa immaculata present	integer	"-999" or "0" or "1"
Anguilla anguilla historic	integer	"-999" or "0" or "1"
Anguilla anguilla present	integer	"-999" or "0" or "1"
Huso huso historic	integer	"-999" or "0" or "1"
Huso huso present	integer	"-999" or "0" or "1"
Lampetra fluviatilis historic	integer	"-999" or "0" or "1"
Lampetra fluviatilis present	integer	"-999" or "0" or "1"
Petromyzon marinus historic	integer	"-999" or "0" or "1"
Petromyzon marinus present	integer	"-999" or "0" or "1"
Osmerus eperlanus historic	integer	"-999" or "0" or "1"
Osmerus eperlanus present	integer	"-999" or "0" or "1"
Salmo salar historic	integer	"-999" or "0" or "1"
Salmo salar present	integer	"-999" or "0" or "1"
Salmo trutta trutta historic	integer	"-999" or "0" or "1"
Salmo trutta trutta present	integer	"-999" or "0" or "1"
Platichthys flesus historic	integer	"-999" or "0" or "1"
Platichthys flesus present	integer	"-999" or "0" or "1"
Coregonus spp. diadr. form historic	integer	"-999" or "0" or "1"
Coregonus spp. diadr. form present	integer	"-999" or "0" or "1"

For the R application, text format seems an acceptable compromise to exchange the data between php application and R application. The structure of the R application is presented in the ***Figure 2***. There are certainly more optimal solutions, but if we consider the time constraints, our actual solution is acceptable.

Root	First folder level	Second folder level	Comments
/efitools	/input	input.txt diadromous.txt	Text files (separator is a **semi-colon**).
	/ouput	output.txt	Text file (separator is a **semi-colon**).
	/internal	models.rda lengthmodels.rda geomodels.rda guild.rda stdpacklength.rda stdpackTrout.rda stdpackNoTrout.rda qclassMean.rda efiminus.R	Internal objects contain in the file called 'internal.zip' and Additional R functions (efiminus.R)
	/man	efitools.pdf	documentation
	maindiad.R		main function with computation of diadromous metric
	main.R		main function without the computation of diadromous metric

Figure 2. Structure of the R application (efitools version 1.0).

3.2- Installation

1- Create a new directory:

Example:

D:/efitools-0.2 **or** D:/programs/efitools-0.2

2- Unzip and keep the folder structure of the archive

3- In the file 'main.R' (and in the file 'maindiad.R'), you must indicate the directory which contains the application efitools (see lines 74-80).

Example :

directory <- " D:/efitools-1.0" **or** directory <- " D:/programs/efitools-1.0"

3.3- How to use R-script to compute EFI index

The command line to perform the index computation is given by the following form:

```
R --vanilla --slave --args “input.txt” “output.txt” < main.R >
info.out
```

The first argument “input.txt” corresponds to the file which contains the initial data from the checking procedure. The second argument “output.txt” is the name of the file which will contain the observed and expected values, individual scores by metric and final score. The file main.R is the main function to compute all the elements necessary to compute scores and indices. The last file called “info.out” contains running information. If there are not any warnings and error messages, the file contains the running time. Otherwise the file contains the specific warning and errors messages.

In this new version (EFItools-1.0), the computation of the metric based on diadromous species is enabled. The command line is similar to the previous one, but we use the function maindiad.R and we indicate the name of the file which contains the information on the diadromous species. The command line corresponds to the following instruction:

```
R --vanilla --slave --args “input.txt” “diadromous.txt” “output.txt”
< maindiad.R > info.out
```

to sum up …

If the file called 'diadromous.txt' exists, you use the function called 'maindiad.R', and if this one doesn't exist, we use simply the function called 'main.R'.

3.4- Description of the R application (file main.R or maindiad.R)

Here is a quick and simplified description of the file **main.R** (or maindiad.R) used to compute the EFI plus indices. The main step is described in the following list:

1- Directory is initially fixed by the programmer (lines 74 – 80).

Example: directory <- "H:/DepotSVN/SVNefiminus/trunk/efitools-1.0"

2- Application loads the necessary objects to compute the indices (lines 85-103).
3- Checking the names of the input and output files (lines 111-126)
4- Importation of input files and additional checking (lines 131-166)
5- Data preparation and computation of the observed metrics (lines 171-217
6- Computation of the expected metrics (lines 222-223)
7- Computation of the individual score (for each separated metric, lines 238-248)
8- First output preparation and table aggregations (Lines 253-280)
9- Metric Aggregation and computation of the indices (for trout and cyprinids zones and global index, line 288).
10- Recoding the fields of the output files and write it in the file “output.txt”(Lines 310-321)
11- End of the R application (return the time of execution). The R application provides directly the output file in text format without intermediate files.
12- Computational events and information (ex. error message, time, etc) are contained in the file called “info.out”.

3.5- Output file

The description of the variables contained in the output file is described in the Table 3. In this file, we conserve the primary key values (Site, Date, Longitude, Latitude) and we add the new values provided by the R application: specific regionalisation variables, observed values, expected values[2]. Presently, the computation of the metric based on the diadromous species is not yet implemented.

[2] **NOTA: Terminology to name the prediction**

Table 2. Description and structure of output file from R application. *The change with the previous version is highlighted in Yellow.*

variables	type	Description
Sample.code	Character	Site name
Day	Integer	e.g.: 08
Month	Integer	e.g.: 10
Year	Integer	e.g.: 2008
Longitude	Float	Longitude (in decimal degree)
Latitude	Float	Latitude (in decimal degree)
Obs.dens.HINTOL.inf.150	Float	Observed number of fish inferior or equal to 150mm and intolerant to habitat degradation by 100 m2
Obs.dens.O2INTOL	Float	Observed number of fish intolerant to low concentration in O2 by 100 m2
Obs.ric.RH.PAR	Float	Observed number of species with preference to spawn in running waters
Obs.dens.LITH	Float	Observed number of lithophilic fish by 100 m2
Exp.dens.HINTOL.inf150	Float	Expected number of fish inferior or equal to 150mm and intolerant to habitat degradation by 100 m2
Exp.dens.O2INTOL	Float	Expected number of fish intolerant to low concentration in O2 by 100 m2
Exp.ric.RH.PAR	Float	Expected number of species with preference to spawn in running waters
Exp.dens.LITH	Float	Expected number of lithophilic fish by 100 m2
Ids.dens.HINTOL.inf.150	Float	Individual score based on number of fish inferior or equal to 150mm and intolerant to habitat degradation by 100 m2
Ids.dens.O2INTOL	Float	Individual score based on number of fish intolerant to low concentration in O2 by 100 m2
Ids.ric.RH.PAR	Float	Individual score based on number of species with preference to spawn in running waters
Ids.dens.LITH	Float	Individual score based on number of lithophilic fish by 100 m2
Method	Character	Boating/Wading/Mixed
Comment.method	Character	Comments on the validity of the Fish index (related to the sampling method)
Sampling.location	Character	Backwaters/Mixed/Main Channel
Comment.sampling.location	Character	Comment on the validity of fish index (related to sampling location
Richness		Species Number considered in the computational procedure (we only consider species described in the guild table)

I used the term “expected values” to name the predictions, because it’s the classical statistical term (the term “predicted values” could be used to name the predictions). I think that the term ‘reference values” is not adapted because models are based on calibration dataset which contains pure reference and quasi-reference sites (low impacted site). Concerning the term “calculated values”, I think that this term is not accurate, because the “observed values” come from a computational process.

Comments.richness		Comments on the validity of the Fish Index (related to the number of species)
Captures	integer	Fish Number considered in the computational procedure (we only consider species described in the guild table)
Comments.sampling.effort	Character	Comments on the validity of the Fish Index (related to the number of fish caught)
Ecoregion	Character	Regionalisation based on ecoregion and mediterranean type used to compute the scores
EFT.river.typology	Character	Functional typology (“Salmonid” or “Cyprinid”)
ST-Species	Float	Proportion of ST-species (Salmonid type species)
River.zone	Character	Corrected river zone considering EFT-type, % of ST-Species. Most of time to be checked by the end-user
Comment.river.zone	Character	Comments on the validity of river zone and recommendations to the end-user
Aggregated.score.Salmonid.zone	Float	Mean of the corresponding metrics for the considered salmonid river zone
Aggregated.score.Cyprinid.zone	Float	Mean of the corresponding metrics for the considered cyprinid river zone
Fish.Index	Float	Final fish index related the river zone
Fish.Index.class	Integer	Classes of the Fish index related to the river zone and the sampling method

If the diadromous metric is computed, we add the three following additional fields:

variables	type	Description
Hist.ric.diadromous	Integer	Historical number of diadromous fish species recorded at the segment scale
Present.ric.diadromous	Integer	Present number of diadromous fish species occurring at the segment scale
Ids.ric.diadromous	Float	Individual score based on diadromous species richness

3.6- Additional Information and limitation

- In this present version the missing value is coded by the term “NA” (native code for missing value in R).
- The index computation for only one site is presently deprecated, because R has some problem in the manipulation of data.frame (metadata in R) with only one row.

4- Reference

R Development Core Team (2007). R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria. ISBN 3-900051-07-0, URL http://www.R-project.org.